

Major Changes from PostgreSQL 11 to PostgreSQL 18

PostgreSQL 11 (Released October 18, 2018)

Performance and Partitioning

- **Partitioning improvements:** Support for hash partitioning, default partitions, and foreign key support on partitioned tables
- **Partition-wise joins and aggregates:** Improved query performance on partitioned tables
- **Faster partition pruning:** More efficient elimination of unnecessary partitions during query planning

SQL Features

- **Stored procedures with transaction control:** CALL statement and procedures that can manage transactions
- **SQL standard GROUPS and EXCLUDE clauses** for window functions
- **CREATE INDEX CONCURRENTLY** for partitioned tables

Parallel Query Improvements

- Parallel hash joins
- Parallel creation of B-tree indexes
- Parallel append for UNION operations

Just-in-Time (JIT) Compilation

- JIT compilation for expressions using LLVM
- Significant performance improvements for complex queries

PostgreSQL 12 (Released October 3, 2019)

Performance and Partitioning

- **Major partitioning performance improvements:** Tables with thousands of partitions now processed efficiently
- **Foreign keys can reference partitioned tables**
- **Improved COPY performance** into partitioned tables
- **Partition bounds can be any expression**

SQL Features

- **SQL/JSON path expressions:** Query JSON documents using SQL/JSON standard
- **Generated columns:** Support for stored generated columns
- **Common Table Expression (CTE) improvements:** CTEs can now be automatically inlined
- **REINDEX CONCURRENTLY:** Rebuild indexes without blocking writes

Indexing Improvements

- Reduced WAL overhead for GiST, GIN, and SP-GiST indexes
- Support for covering GiST indexes via INCLUDE clause
- Multi-column most-common-value (MCV) statistics via CREATE STATISTICS

Authentication

- GSSAPI encryption support
- Client-side and server-side encryption for authentication

Removed Features

- **WITH OIDS support removed:** Cannot specify OID columns during table creation

PostgreSQL 13 (Released September 24, 2020)

Performance Improvements

- **B-tree index deduplication:** Significant space savings for indexes with duplicate values
- **Incremental sorting:** Better performance for queries with sorted results
- **Parallel VACUUM for indexes:** Faster maintenance operations
- **Improved hash aggregation:** Better memory management and disk spilling

Monitoring and Administration

- **WAL usage statistics in EXPLAIN**
- **Progress reporting** for ANALYZE commands and streaming base backups
- **pg_stat_progress_basebackup** view for monitoring backup progress

Security

- **Channel binding for SCRAM authentication**
- **Trusted extensions:** More secure extension management
- **Default minimum TLS version changed from 1.0 to 1.2**

Removed Features

- Support for pre-PostgreSQL 8.0 operator class syntax
- Support for upgrading unpackaged (pre-9.1) extensions
- Support for OpenSSL 0.9.8 and 1.0.0

PostgreSQL 14 (Released September 30, 2021)

Performance and Concurrency

- **Query pipelining:** Send multiple queries without waiting for results
- **Connection scaling improvements:** Better performance for high connection counts
- **Parallel sequential scan improvements**
- **VACUUM emergency mode:** Prevents transaction ID wraparound

Logical Replication

- **Streaming of in-progress transactions** to subscribers
- **Binary format support** for logical replication
- **Progress tracking** for logical replication

SQL and Procedural Languages

- **SQL-standard SEARCH and CYCLE clauses** for recursive CTEs
- **Multirange data types:** Store non-contiguous ranges
- **Subscripting for jsonb** using bracket notation
- **OUT parameters for procedures**

Administration

- **pg_read_all_data and pg_write_all_data roles:** Simplified permission management
- **pg_amcheck utility:** Database corruption checking
- **REINDEX improvements:** Better performance and progress reporting

Monitoring

- **Query ID tracking** with `compute_query_id`
- **COPY command progress** tracking
- **Enhanced statistics** for query planning

PostgreSQL 15 (Released October 13, 2022)

SQL Features

- **MERGE command:** SQL-standard UPSERT-like functionality
- **SQL/JSON constructors:** JSON_TABLE, JSON_VALUE, JSON_QUERY
- **Regular expression functions:** regexp_count(), regexp_instr(), regexp_like(), regexp_substr()

Performance

- **Sorting improvements:** Up to 4x faster in some cases
- **Compression improvements:** Better WAL compression with LZ4 and Zstandard
- **SELECT DISTINCT optimization:** Significant performance improvements

Logical Replication

- **Row filtering and column lists** for publishers
- **Two-phase commit support** for logical replication
- **Publishers support all table types** including partitioned tables
- **Conflict management:** Skip conflicting transactions, auto-disable on errors

Security and Administration

- **PUBLIC creation permission removed** from public schema (security hardening)
- **Collation version tracking:** Detect and prevent index corruption
- **jsonlog format:** Structured logging in JSON format
- **ICU collations** can be set as cluster/database defaults

Removed Features

- Exclusive backup mode completely removed
- Python 2 support removed from PL/Python

PostgreSQL 16 (Released September 14, 2023)

Performance

- **Query parallelism improvements:** Parallel FULL and RIGHT joins
- **Incremental sorts for aggregates:** Better memory usage
- **SIMD CPU acceleration:** For ASCII and JSON string processing
- **I/O improvements:** Better read/write patterns

Logical Replication

- **Bidirectional logical replication** support
- **Logical replication from standby servers**
- **Parallel apply** for large transactions

Monitoring

- **pg_stat_io view:** Detailed I/O statistics
- **Granular VACUUM control:** More options for maintenance
- **Load balancing in libpq:** Multiple host connection support

Developer Features

- **SQL/JSON constructors and identity functions**
- **ICU collation improvements:** Can be set as default
- **ANY_VALUE aggregate function**
- **EXPLAIN enhancements:** More detailed buffer and I/O information

PostgreSQL 17 (Released September 19, 2024)

Performance

- **2x improvement in VACUUM memory management**
- **Improved bitmap heap scan performance**
- **Better IN/NOT IN query performance** with B-tree indexes
- **Incremental backup support:** Reduced backup time and storage

SQL Features

- **JSON_TABLE function:** Convert JSON to relational data
- **COPY ON_ERROR option:** Continue copying despite errors
- **EXPLAIN improvements:** Shows memory usage

Storage and I/O

- **Incremental backup** with pg_basebackup
- **Improved WAL performance:** Better I/O patterns
- **Block-level incremental backup** capabilities

Administration

- **pg_maintain predefined role:** For maintenance operations
- **pg_wait_events system view:** Detailed wait event documentation
- **Enhanced SSL/TLS configuration** options

PostgreSQL 18 (Released September 25, 2025)

Asynchronous I/O (AIO) - Revolutionary Change

- **New AIO subsystem:** Up to 3x performance improvement for reads
- **io_method parameter:** sync, worker (default), or io_uring (Linux)
- **Supports:** Sequential scans, bitmap heap scans, vacuums
- **pg_aios system view:** Monitor async I/O operations

Developer Features

- **Virtual generated columns:** Now the default (compute at read time)
- **uuidv7() function:** Timestamp-ordered UUIDs for better indexing

Enhanced RETURNING clause: Support for OLD and NEW values

UPDATE accounts SET balance = balance - 50 RETURNING OLD.balance, NEW.balance;

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Query Performance

- **Skip scan:** Use multicolumn indexes without all leading columns
- **Improved OR/IN processing:** Automatic index optimization
- **Parallel GIN index builds**
- **Better hash join performance**

Security & Authentication

- **OAuth 2.0 authentication support:** Easy SSO integration
- **Data checksums enabled by default:** Better data integrity
- **MD5 authentication deprecated:** Will be removed in future
- **Wire protocol 3.2:** First update since 2003

Schema Management

NOT NULL constraints as NOT VALID: Add without table scan

```
ALTER TABLE large_table ADD CONSTRAINT check_email CHECK (email IS NOT NULL)
NOT VALID;-- Validate later with minimal lockingALTER TABLE large_table VALIDATE
CONSTRAINT check_email;
```

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- **Temporal constraints:** Range-based PRIMARY KEY, UNIQUE, FOREIGN KEY
- **NOT ENFORCED constraints:** Deferred enforcement option

Upgrade & Maintenance

- **Statistics preserved during upgrades:** Faster post-upgrade performance
- **pg_upgrade improvements:** Parallel checks (--jobs), swap directories (--swap)

- **Better VACUUM behavior:** Adaptive aggressive vacuum, improved cost tracking

Text Processing

- **PG_UNICODE_FAST collation:** Fast Unicode with full semantics
- **casefold() function:** Case-insensitive comparisons
- **LIKE with nondeterministic collations:** More flexible pattern matching

Observability

- **EXPLAIN ANALYZE includes buffer usage by default**
- **pg_get_acl() function:** Inspect object privileges
- **Enhanced monitoring:** track_cost_delay_timing for vacuum

Summary of Major Themes (PostgreSQL 11-18)

Performance Evolution

- **PostgreSQL 11:** JIT compilation introduction
- **PostgreSQL 12-14:** Partitioning and parallel query improvements
- **PostgreSQL 15-17:** Sorting and I/O optimizations
- **PostgreSQL 18:** Revolutionary async I/O (3x read performance)

Developer Experience

- **SQL Standards:** MERGE, SQL/JSON, temporal constraints
- **Data Types:** Generated columns, multirange, UUID v7
- **Query Features:** Enhanced RETURNING, skip scans

Replication & High Availability

- **Evolution:** Basic → Filtered → Bidirectional → Parallel apply
- **Flexibility:** Row/column filtering, standby replication, conflict management

Security & Authentication

- **Modern Auth:** OAuth 2.0, SCRAM improvements
- **Data Integrity:** Checksums by default, collation version tracking
- **Deprecations:** MD5 authentication, Python 2

Operational Excellence

- **Monitoring:** pg_stat_io, wait events, progress tracking
- **Maintenance:** VACUUM improvements, incremental backups
- **Upgrades:** Statistics preservation, faster pg_upgrade

Key Deprecations/Removals Timeline

- **PostgreSQL 12:** WITH OIDS removed
- **PostgreSQL 13:** OpenSSL 0.9.8/1.0.0 support removed
- **PostgreSQL 15:** Exclusive backup, Python 2 removed
- **PostgreSQL 18:** MD5 authentication deprecated (removal pending)

This document tracks the evolution of PostgreSQL from version 11 (2018) through version 18 (2025), highlighting the continuous improvements in performance, developer experience, security, and operational capabilities.